

Volume By Rotation

Section 7.2

1-16-2026

A) Line Rotations

Horizontal Line Rotation Vertical Line Rotation

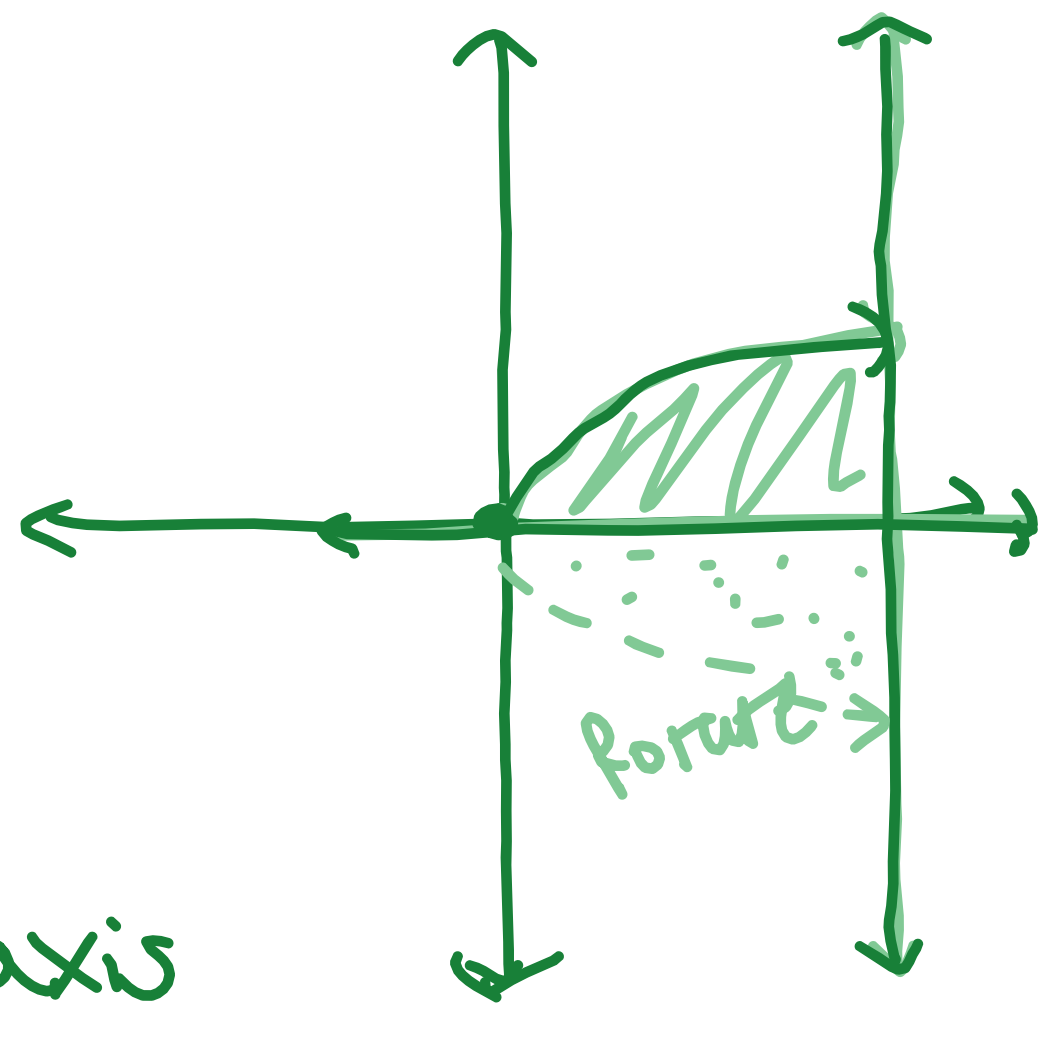
$$\pi \int_a^b r^2 dx$$

$$\pi \int_c^d r^2 dy$$

B) Base

Ex:

$$\begin{cases} y = \sqrt{x} \\ x = 4 \\ y = 0 \end{cases}$$

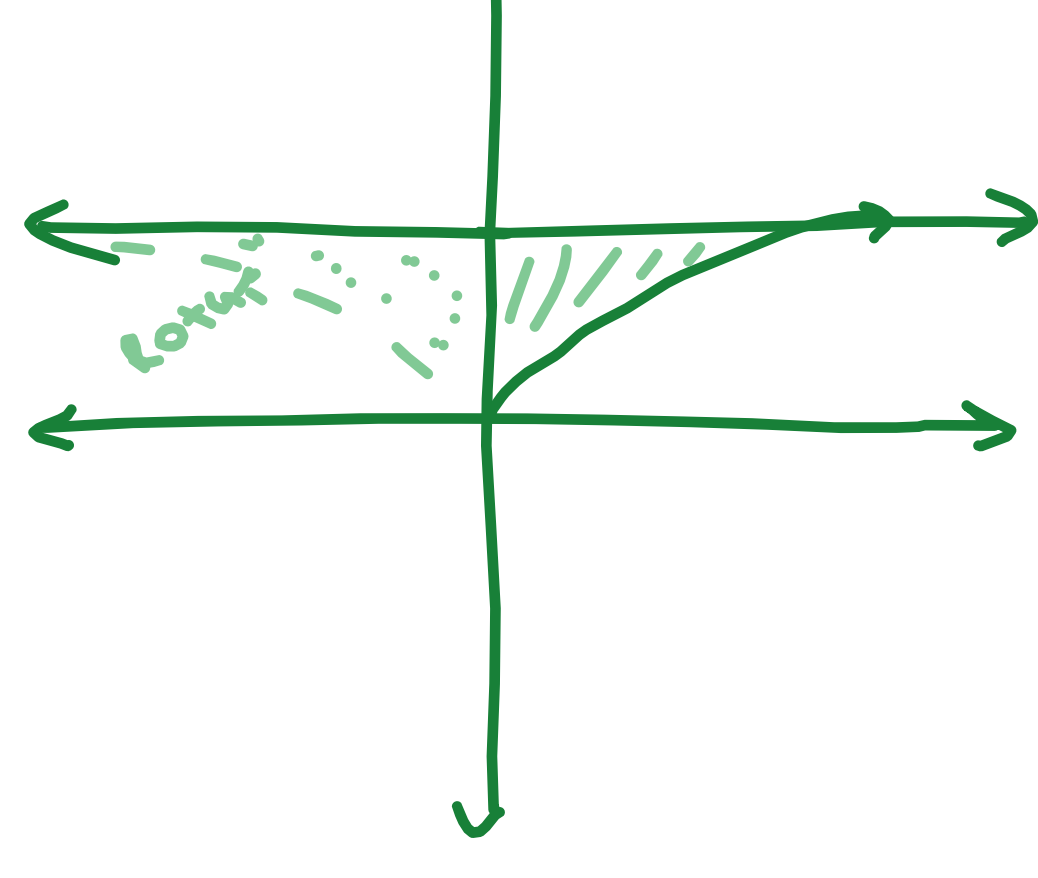


Rotate on x-axis

$$\pi \int_0^4 (\sqrt{x} - 0)^2 dx$$

Ex:

$$\begin{cases} y = \sqrt{x} \\ y = 2 \\ x = 0 \end{cases}$$

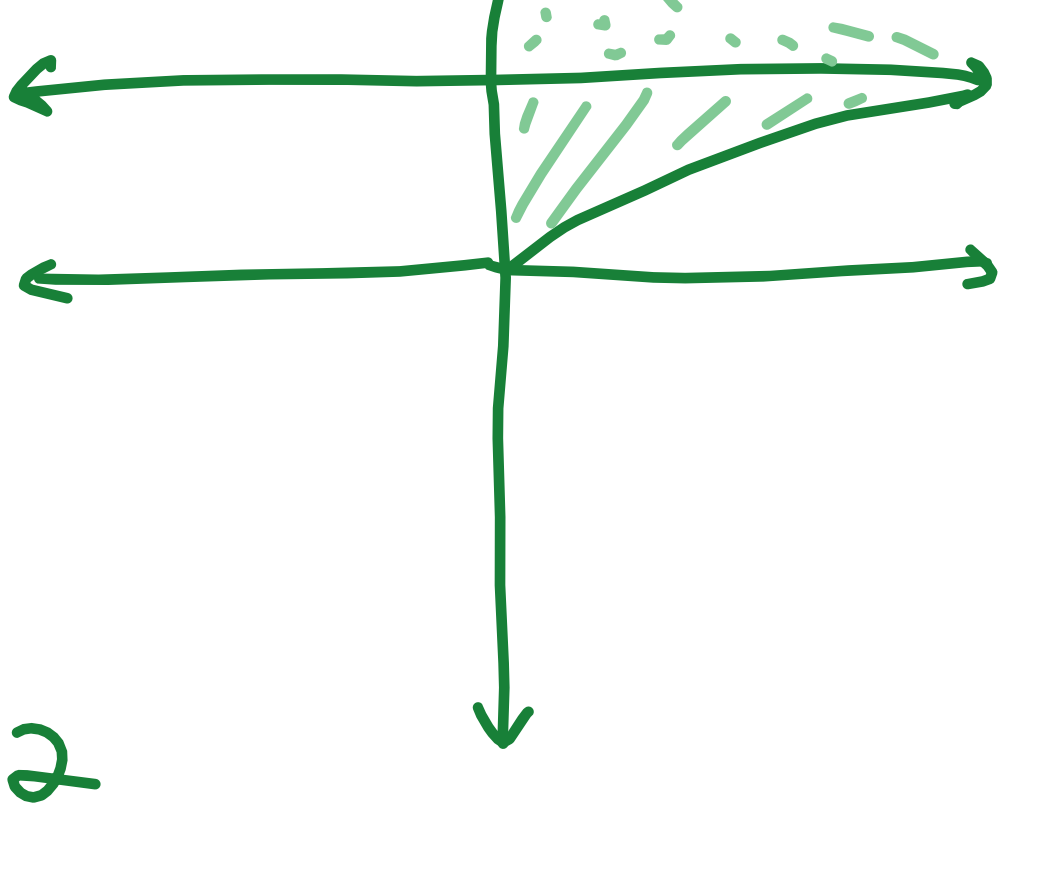


Rotate y-axis

$$\pi \int_0^2 (y^2 - 0)^2 dy$$

Ex:

$$\begin{cases} y = \sqrt{x} \\ y = 2 \\ x = 0 \end{cases}$$

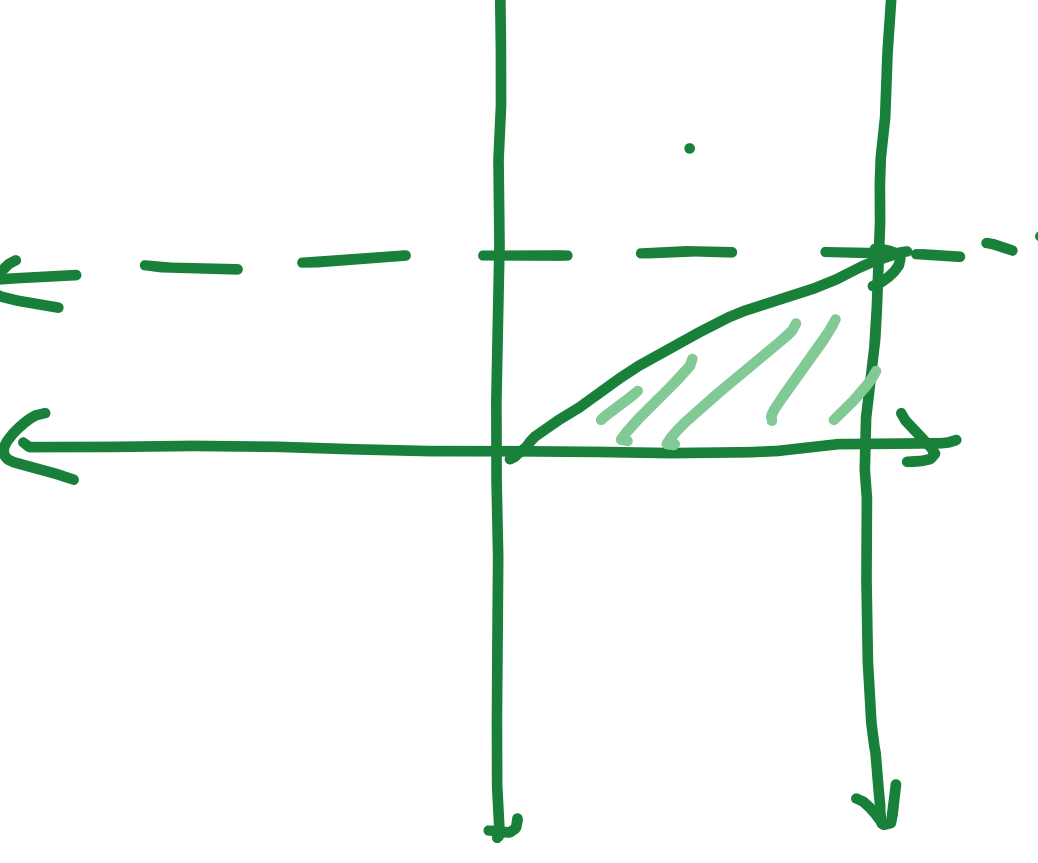


Rotate y=2

$$\pi \int_0^4 (2 - \sqrt{x})^2 dx$$

Ex:

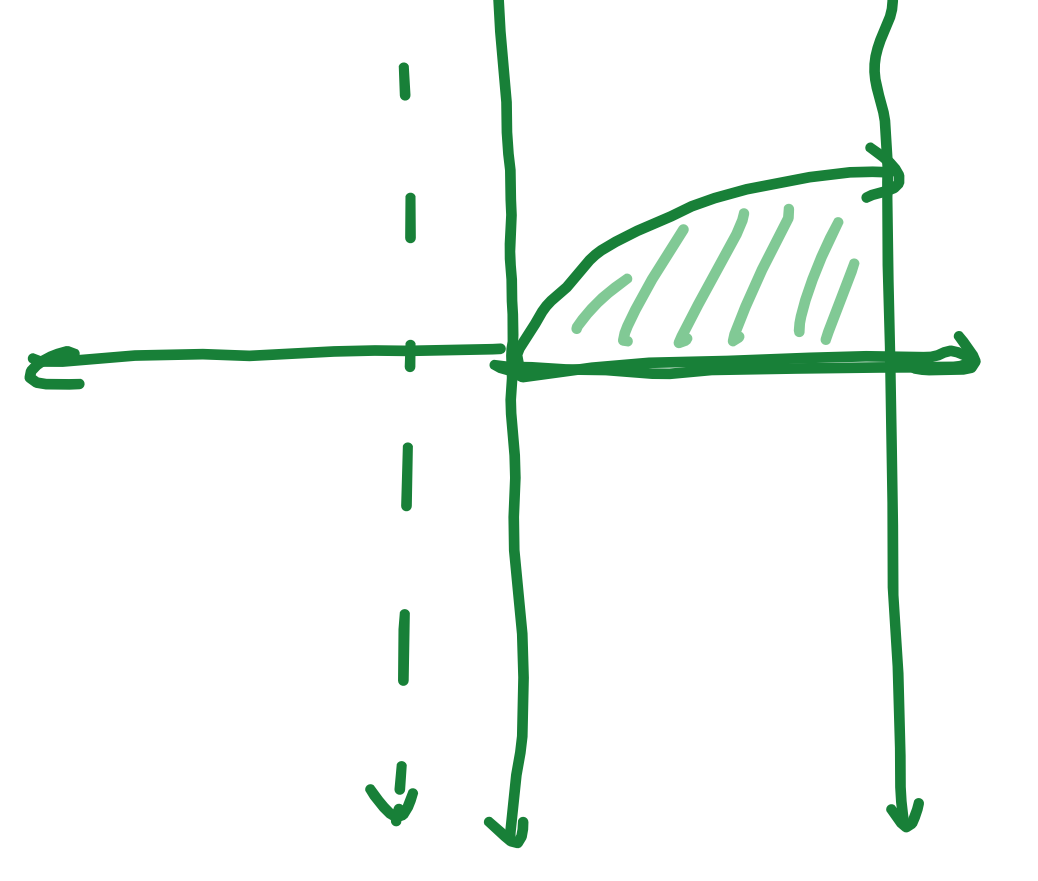
$$\begin{cases} y = \sqrt{x} \\ x = 4 \\ x = 0 \end{cases}$$



Rotate y=2

$$\pi \int_0^4 (2)^2 - (2 - \sqrt{x})^2 dx$$

$$\begin{cases} y = \sqrt{x} \\ x = 4 \\ y = 0 \end{cases}$$

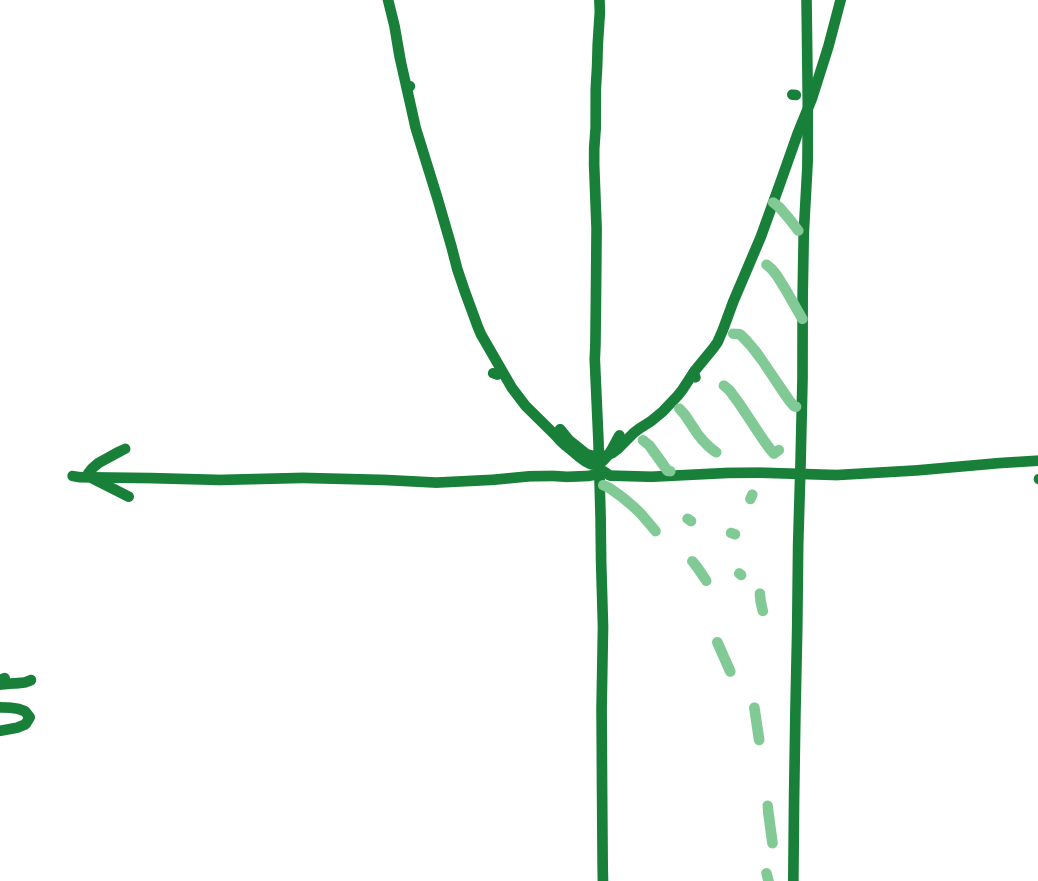


Rotate x=-1

$$\pi \int_0^2 (4 + 1)^2 - (y^2 + 1)$$

Ex:

$$\begin{cases} y = x^2 \\ x = 2 \\ y = 0 \end{cases}$$



a) Rotate y=0

b) Rotate x=5

$$a) \pi \int_0^2 (x^2)^2 dx$$

$$b) \pi \int_0^4 (5 - \sqrt{y})^2 - (3)^2 dy$$